BCA EXAMINATION NOTES

Course Code: CA-124

CA-124 Basic Mathematics for Computing

Exam Pattern:

Q1. Attempt Any Six – 6 marks

Q2. Attempt Any Three – 6 marks

Q3. Attempt Any Two – 6 marks

Q4A. Attempt Any One – 4 marks | Q4B. Compulsory – 2 marks

Q5. Attempt Any One – 6 marks

UNIT 1: Mathematical Logic

1.1 Meaning of Statement

A statement (proposition) is a declarative sentence that is either TRUE or FALSE, but not both. Statements have a definite truth value.

Examples of statements: 'The sun rises in the east' (TRUE), '2+2=5' (FALSE)

Non-statements: Questions ('Is it raining?'), commands ('Close the door'), exclamations

1.2 Primitive and Compound Statements

Primitive (Simple) Statement

A statement that cannot be broken down into simpler statements. It is the basic unit. Denoted by p, q, r, etc.

Example: p: 'It is raining.' q: 'It is cold.'

Compound Statement

A statement formed by combining two or more primitive statements using logical connectives (and, or, not, if-then, if-and-only-if).

Example: 'It is raining AND it is cold.' — $p \wedge q$

1.3 Truth Values

Every statement has a truth value: T (True) or F (False). The truth value of a compound statement depends on the truth values of its components and the connective used.

1.4 Logical Operations (Connectives)

- **Negation (NOT)** ($\neg p$ or $\sim p$): If p is T, $\neg p$ is F; if p is F, $\neg p$ is T
- **Conjunction (AND)** ($p \wedge q$): True only when both p and q are True
- **Disjunction (OR)** ($p \vee q$): True when at least one of p, q is True
- **Conditional (IF-THEN)** ($p \rightarrow q$): False only when p is True and q is False
- **Biconditional (IFF)** ($p \leftrightarrow q$): True when both p and q have same truth value

1.5 Truth Tables

A truth table systematically shows all possible truth value combinations of component statements and the resulting truth value of the compound statement.

For n variables, there are 2^n rows in the truth table.

Example Truth Table for $p \wedge q$:

p	q	$p \wedge q$	$p \vee q$	$p \rightarrow q$	$p \leftrightarrow q$
T	T	T	T	T	T
T	F	F	T	F	F
F	T	F	T	T	F
F	F	F	F	T	T

1.6 Equivalence of Logical Statements

Two statements P and Q are logically equivalent if they have the same truth value for all possible combinations. Written as $P \equiv Q$ or P ■ Q.

Important Equivalences (Laws):

- Double Negation: $\neg(\neg p) \equiv p$
- De Morgan's Laws: $\neg(p \wedge q) \equiv \neg p \vee \neg q$ | $\neg(p \vee q) \equiv \neg p \wedge \neg q$
- Commutative: $p \wedge q \equiv q \wedge p$ | $p \vee q \equiv q \vee p$
- Associative: $(p \wedge q) \wedge r \equiv p \wedge (q \wedge r)$
- Distributive: $p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$
- Identity: $p \wedge T \equiv p$ | $p \vee F \equiv p$
- Idempotent: $p \wedge p \equiv p$ | $p \vee p \equiv p$

1.7 Tautology and Contradiction

Tautology: A compound statement that is always TRUE regardless of the truth values of its components. Example: $p \vee \neg p$ (always True)

Contradiction (Fallacy): A compound statement that is always FALSE. Example: $p \wedge \neg p$ (always False)

Contingency: A statement that is neither a tautology nor a contradiction.

UNIT 2: Sets

2.1 Meaning of a Set

A set is a well-defined collection of distinct objects called elements or members. 'Well-defined' means we can clearly determine if an object belongs to the set or not.

Notation: Sets are denoted by capital letters (A, B, C). Elements by small letters (a, b, c). Membership: $a \in A$ (a belongs to A), $b \notin A$ (b does not belong to A).

2.2 Methods of Describing a Set

Tabular (Roster) Form

List all elements inside curly braces, separated by commas.

Example: $A = \{1, 2, 3, 4, 5\}$ — Set of first five natural numbers

Set Builder Form

Describe the property that defines the set.

Example: $A = \{x \mid x \text{ is a natural number and } x \leq 5\}$ — same set as above

2.3 Types of Sets

Finite Set: Has countable number of elements. E.g., $A = \{1,2,3\}$

Infinite Set: Has unlimited elements. E.g., $N = \{1,2,3,...\}$

Empty Set (Null Set): Has no elements. Denoted by {} or ϕ. E.g., Set of even primes between 10 and 20

Subset: A is subset of B ($A \subseteq B$) if every element of A is in B. Every set is subset of itself.

Universal Set (U): Contains all sets under consideration in a given context.

Equal Sets: $A = B$ if they have exactly the same elements.

Overlapping Sets: Have at least one element in common but neither is subset of other.

Disjoint Sets: Have no elements in common. $A \cap B = \phi$

Complementary Set: $A' = U - A$. Contains all elements of U not in A.

2.4 Operations on Sets

Union ($A \cup B$)

Set of all elements in A or B or both. $A \cup B = \{x \mid x \in A \text{ or } x \in B\}$

Intersection ($A \cap B$)

Set of all elements common to both A and B. $A \cap B = \{x \mid x \in A \text{ and } x \in B\}$

Difference (A - B)

Set of elements in A but not in B. $A - B = \{x \mid x \in A \text{ and } x \notin B\}$

Cartesian Product ($A \times B$)

Set of all ordered pairs (a,b) where $a \in A$ and $b \in B$. $n(A\times B) = n(A) \times n(B)$

Example: A={1,2}, B={a,b} $\rightarrow$ $A\times B = \{(1,a),(1,b),(2,a),(2,b)\}$

2.5 Venn Diagrams

Venn diagrams are pictorial representations of sets using circles within a rectangle (universal set). Overlapping circles show common elements.

Used to illustrate: Union (total area), Intersection (overlap), Difference (one circle excluding overlap), Complement (area outside circle).

UNIT 3: Determinants

3.1 Meaning and Properties of Determinants

A determinant is a scalar value computed from a square matrix. Denoted |A| or det(A).

Second order: |A| = |a b; c d| = ad - bc

Properties:

- If any row/column is all zeros, determinant = 0
- Interchanging two rows/columns changes sign of determinant
- If two rows/columns are identical, determinant = 0
- Multiplying a row/column by k multiplies determinant by k
- Adding a multiple of one row to another does not change determinant
- det(A) = det(A^T) (transpose property)
- det(AB) = det(A) × det(B)

3.2 Evaluation of Second Order Determinants

For 2×2 matrix A = [[a,b],[c,d]]:

|A| = ad - bc

Example: |[3,2],[1,4]| = (3×4) - (2×1) = 12 - 2 = 10

3.3 Minor and Cofactor

Minor of element a_ij: The determinant of the matrix obtained by deleting row i and column j. Denoted M_ij.

Cofactor of element a_ij: A_ij = (-1)^(i+j) × M_ij

The sign factor (-1)^(i+j) gives: + for even (i+j), - for odd (i+j). Pattern: +,-,+; -,+,-; +,-,+

3.4 Cramer's Rule

Cramer's rule solves systems of linear equations using determinants.

For 2 equations: $a_1x+b_1y=c_1$ and $a_2x+b_2y=c_2$

$D = |a_1\ b_1;\ a_2\ b_2| = a_1b_2 - a_2b_1$

$D_x = |c_1\ b_1;\ c_2\ b_2| = c_1b_2 - c_2b_1$

$D_y = |a_1\ c_1;\ a_2\ c_2| = a_1c_2 - a_2c_1$

Solution: $x = D_x/D$, $y = D_y/D$ (provided $D \neq 0$)

UNIT 4: Matrices

4.1 Meaning and Order of Matrix

A matrix is a rectangular arrangement of numbers in rows and columns enclosed in square brackets [ ] or parentheses ( ).

Order: A matrix with m rows and n columns is called an $m \times n$ matrix. Element in row i, column j is denoted a_{ij}.

4.2 Types of Matrix

- **Row Matrix:** Only one row. Order: $1 \times n$
- **Column Matrix:** Only one column. Order: $m \times 1$
- **Square Matrix:** Number of rows = number of columns ($m=n$)
- **Zero Matrix (Null):** All elements are 0
- **Identity Matrix:** Square matrix with 1s on diagonal and 0s elsewhere
- **Diagonal Matrix:** Square matrix with non-zero elements only on main diagonal
- **Symmetric Matrix:** $A = A^T$ (element $a_{ij} = a_{ji}$)
- **Upper Triangular:** All elements below main diagonal are 0
- **Lower Triangular:** All elements above main diagonal are 0

4.3 Algebra of Matrices

Equality

Two matrices A and B are equal if they have the same order and $a_{ij} = b_{ij}$ for all i,j.

Scalar Multiplication

Multiplying matrix A by scalar k: each element is multiplied by k. $kA = [k \cdot a_{ij}]$

Addition of Matrices

$A + B$ is defined only when A and B have same order. Add corresponding elements: $(A+B)_{ij} = a_{ij} + b_{ij}$

Subtraction

$A - B = A + (-B)$ where $-B = (-1)B$. Subtract corresponding elements.

Multiplication of Matrices

$A\ (m \times n) \times B\ (n \times p) = C\ (m \times p)$. c_{ij} = sum of $a_{ik} \times b_{kj}$ for k=1 to n.

Number of columns of A must equal number of rows of B. Matrix multiplication is NOT commutative ($AB \neq BA$ in general).

DEFINITIONS (2-Mark Ready)

Statement: A statement is a declarative sentence that has a definite truth value, either TRUE or FALSE, but not both.

Tautology: A tautology is a compound logical statement that is always TRUE, regardless of the truth values of its component statements.

Contradiction: A contradiction is a compound logical statement that is always FALSE for all possible truth value combinations.

Set: A set is a well-defined collection of distinct objects called elements, denoted by capital letters and listed within curly braces.

Subset: Set A is a subset of set B ($A \subseteq B$) if every element of A is also an element of B.

Determinant: A determinant is a scalar value associated with a square matrix, computed from its elements using a defined formula.

Minor: The minor of element a_ij in a matrix is the determinant of the matrix obtained by deleting row i and column j.

Cofactor: The cofactor of a_ij is (-1)^(i+j) times the minor M_ij, accounting for the sign based on position.

Matrix: A matrix is a rectangular array of numbers arranged in m rows and n columns, enclosed in brackets, with order m×n.

Identity Matrix: An identity matrix is a square matrix with 1s on the main diagonal and 0s elsewhere, acting as multiplicative identity.

20 MOST IMPORTANT QUESTIONS WITH ANSWERS

Q1. Define statement. Give examples of statements and non-statements. [2M]

A statement is a declarative sentence with a definite truth value (TRUE or FALSE).
Statements: '5 is prime' (T), 'Earth has two moons' (F), '7+3=10' (T)
Non-statements: 'Is it cold?' (question), 'Close the door' (command), 'How beautiful!' (exclamation)
A sentence is a statement only if we can assign it a truth value.

Q2. Explain logical connectives with symbols and truth values. [4M]

1. NEGATION (¬p): Reverses truth value. If p=T, ¬p=F. If p=F, ¬p=T.
2. CONJUNCTION ($p \wedge q$): AND. True only when both p and q are True.
3. DISJUNCTION ($p \vee q$): OR. True when at least one is True.
4. CONDITIONAL ($p \rightarrow q$): IF-THEN. False only when p=T and q=F.
$p \rightarrow q$ is equivalent to $\neg p \vee q$.
5. BICONDITIONAL ($p \leftrightarrow q$): IF AND ONLY IF. True when p and q have same truth value.
$p \leftrightarrow q$ is equivalent to $(p \rightarrow q) \wedge (q \rightarrow p)$.

Q3. Construct truth table for $(p \wedge q) \vee \neg r$. [4M]

Truth table with 3 variables has 2^3 = 8 rows:
p q r | p∧q | ¬r | (p∧q)∨¬r
T T T | T | F | T
T T F | T | T | T

T F T | F | F | F
T F F | F | T | T
F T T | F | F | F
F T F | F | T | T
F F T | F | F | F
F F F | F | T | T
Method: First fill p,q,r columns. Compute p∧q. Compute ¬r. Then compute final column.

Q4. What is tautology? Prove p ∨ ¬p is a tautology. [4M]

Tautology: A compound statement that is always TRUE for all possible truth values of variables.
Proof that p ∨ ¬p is a tautology:
p | ¬p | p ∨ ¬p
T | F | T
F | T | T
Since p ∨ ¬p is TRUE for both values of p, it is a tautology.
This is called the Law of Excluded Middle.

Similarly, p ∧ ¬p is a contradiction (always FALSE):.
p | ¬p | p ∧ ¬p
T | F | F
F | T | F

Q5. State and prove De Morgan's laws. [6M]

De Morgan's Laws:
Law 1: ¬(p ∧ q) ≡ ¬p ∨ ¬q
Law 2: ¬(p ∨ q) ≡ ¬p ∧ ¬q

Proof of Law 1 (Truth Table):
p q | p∧q | ¬(p∧q) | ¬p | ¬q | ¬p∨¬q
T T | T | F | F | F | F
T F | F | T | F | T | T
F T | F | T | T | F | T
F F | F | T | T | T | T
Columns ¬(p∧q) and ¬p∨¬q are identical → PROVED.

Proof of Law 2 similarly:
p q | p∨q | ¬(p∨q) | ¬p | ¬q | ¬p∧¬q
T T | T | F | F | F | F
T F | T | F | F | T | F
F T | T | F | T | F | F
F F | F | T | T | T | T
Columns match → PROVED.

Q6. What is a set? Explain different types of sets with examples. [6M]

A set is a well-defined collection of distinct objects.

Types:
1. FINITE SET: Countable elements. A = {1,2,3,4,5}
2. INFINITE SET: Uncountable elements. N = {1,2,3,...}

3. EMPTY SET: No elements. $\phi = \{\}$ e.g., set of even primes >10
4. SINGLETON SET: Exactly one element. $\{5\}$
5. UNIVERSAL SET (U): Contains all relevant sets.
6. EQUAL SETS: A = B if same elements. $\{1,2,3\} = \{3,1,2\}$
7. SUBSET: $A \subseteq B$ if all elements of A are in B.
8. PROPER SUBSET: $A \subset B$ if $A \subseteq B$ and $A \neq B$.
9. DISJOINT SETS: $A \cap B = \phi$. No common elements.
10. OVERLAPPING SETS: Have at least one common element.
11. COMPLEMENTARY SET: A' = U - A

Q7. Explain set operations with examples. [6M]

Given: $A = \{1,2,3,4,5\}$, $B = \{3,4,5,6,7\}$, $U = \{1,2,...,8\}$

1. UNION ($A \cup B$): All elements in A or B.
$A \cup B = \{1,2,3,4,5,6,7\}$

2. INTERSECTION ($A \cap B$): Common elements.
$A \cap B = \{3,4,5\}$

3. DIFFERENCE (A - B): In A but not B.
$A - B = \{1,2\}$
$B - A = \{6,7\}$

4. COMPLEMENT (A'): Not in A.
$A' = U - A = \{6,7,8\}$

5. CARTESIAN PRODUCT: $A\times B = \{(a,b) \mid a\in A, b\in B\}$
If $A=\{1,2\}$, $B=\{a,b\}$: $A\times B=\{(1,a),(1,b),(2,a),(2,b)\}$
$n(A\times B) = n(A) \times n(B) = 2\times 2 = 4$

6. SYMMETRIC DIFFERENCE: A■B = $(A-B)\cup(B-A) = \{1,2,6,7\}$

Q8. Explain Venn diagrams with examples. [4M]

Venn diagrams use circles within a rectangle (universal set U) to represent sets visually.

Components:
- Rectangle: Universal set U
- Circles: Individual sets
- Overlapping region: Intersection

Representation of operations:
- Union $A\cup B$: Shaded area = entire both circles
- Intersection $A\cap B$: Shaded area = only overlapping region
- A-B: Only left circle excluding overlap
- A': Everything in rectangle except circle A

Example: Given $A=\{1,2,3\}$, $B=\{3,4,5\}$, $U=\{1,2,3,4,5,6\}$
$A\cap B = \{3\}$ — small overlapping region
$A\cup B = \{1,2,3,4,5\}$ — combined area of both circles

A' = {4,5,6} — outside circle A

Q9. Evaluate 2nd order determinant and explain properties. [4M]

Second Order Determinant:
For A = [[a,b],[c,d]], |A| = ad - bc

Examples:
|3 2|
|1 4| = (3)(4)-(2)(1) = 12-2 = 10

|5 -3|
|2 4| = (5)(4)-(-3)(2) = 20+6 = 26

Properties:
1. |A^T| = |A| (transpose doesn't change value)
2. If any row/column = 0, then |A| = 0
3. Swapping two rows changes sign: |A| → -|A|
4. Two identical rows → |A| = 0
5. kR_i → k|A| (row times k multiplies det by k)
6. Adding multiple of one row to another: |A| unchanged

Q10. Solve using Cramer's Rule: 2x+y=5, 3x-2y=4. [6M]

System: 2x + y = 5 ... (1)
3x - 2y = 4 ... (2)

Step 1: Calculate D (coefficient determinant)
D = |2 1|
|3 -2| = (2)(-2) - (1)(3) = -4-3 = -7

Step 2: Calculate Dx (replace x-column with constants)
Dx = |5 1|
|4 -2| = (5)(-2) - (1)(4) = -10-4 = -14

Step 3: Calculate Dy (replace y-column with constants)
Dy = |2 5|
|3 4| = (2)(4) - (5)(3) = 8-15 = -7

Step 4: Solution
x = Dx/D = -14/-7 = 2
y = Dy/D = -7/-7 = 1

Verification: 2(2)+1=5 ✓, 3(2)-2(1)=4 ✓
Solution: x=2, y=1

Q11. What are types of matrices? [4M]

1. ROW MATRIX: One row, n columns. [1 2 3]
2. COLUMN MATRIX: m rows, one column. [[1],[2],[3]]
3. SQUARE MATRIX: m = n. [[1,2],[3,4]]
4. ZERO/NULL MATRIX: All elements zero. [[0,0],[0,0]]

5. IDENTITY MATRIX: Diagonal 1s, rest 0s. [[1,0],[0,1]]
6. DIAGONAL MATRIX: Non-zero only on diagonal.
7. SYMMETRIC: A = A^T, so a_ij = a_ji for all i,j.
8. SKEW-SYMMETRIC: A = -A^T, diagonal elements = 0.
9. UPPER TRIANGULAR: a_ij=0 when i>j
10. LOWER TRIANGULAR: a_ij=0 when i11. SCALAR MATRIX: Diagonal matrix with equal diagonal elements.

Q12. Explain matrix addition and subtraction with example. [4M]

Matrix Addition: Only possible when both matrices have same order.
(A+B)_ij = a_ij + b_ij (add corresponding elements)

Example:
A = [[1,2,3],[4,5,6]] B = [[7,8,9],[1,2,3]] (both 2×3)
A+B = [[1+7, 2+8, 3+9],[4+1, 5+2, 6+3]]
= [[8,10,12],[5,7,9]]

Matrix Subtraction: A-B = A + (-B)
A-B = [[1-7, 2-8, 3-9],[4-1, 5-2, 6-3]]
= [[-6,-6,-6],[3,3,3]]

Properties:
1. Commutative: A+B = B+A
2. Associative: (A+B)+C = A+(B+C)
3. A+O = A (O is null matrix)
4. A+(-A) = O

Q13. Explain matrix multiplication with example. [6M]

Matrix multiplication A×B is defined when no. of columns of A = no. of rows of B.
If A is m×n and B is n×p, then C = A×B is m×p.
c_ij = sum(a_ik × b_kj) for k=1 to n

Example: A = [[1,2],[3,4]] (2×2), B = [[5,6],[7,8]] (2×2)
C = A×B:
c_11 = 1×5 + 2×7 = 5+14 = 19
c_12 = 1×6 + 2×8 = 6+16 = 22
c_21 = 3×5 + 4×7 = 15+28 = 43
c_22 = 3×6 + 4×8 = 18+32 = 50
C = [[19,22],[43,50]]

Properties:
1. NOT commutative: AB ≠ BA generally
2. Associative: (AB)C = A(BC)
3. Distributive: A(B+C) = AB+AC
4. AI = IA = A (I is identity matrix)
5. If |A| = 0, A has no inverse

Q14. What is scalar multiplication of a matrix? [2M]

Scalar multiplication: When matrix A is multiplied by a scalar (number) k, every element of A is multiplied by k.
If A = [[a,b],[c,d]], then kA = [[ka,kb],[kc,kd]]
Example: A = [[1,2],[3,4]], k=3
3A = [[3,6],[9,12]]
Properties: k(A+B) = kA+kB; (k+l)A = kA+lA; k(lA) = (kl)A

Q15. Prove that $\neg(p \to q) \equiv p \land \neg q$ [4M]

We need to show $\neg(p \to q) \equiv p \land \neg q$
Truth table:
p q | p→q | ¬(p→q) | ¬q | p∧¬q
T T | T | F | F | F
T F | F | T | T | T
F T | T | F | F | F
F F | T | F | T | F
Column $\neg(p \to q)$ and $p \land \neg q$ are identical. PROVED.
Interpretation: 'NOT (if p then q)' is equivalent to 'p is true AND q is false'. The conditional fails only when hypothesis is true but conclusion is false.

Q16. What is Cartesian product of sets? Give example. [4M]

Cartesian Product A × B is the set of all ordered pairs (a,b) where $a \in A$ and $b \in B$.
$A \times B = \{(a,b) \mid a \in A \text{ and } b \in B\}$

Example:
A = {1,2,3}, B = {a,b}
A × B = {(1,a),(1,b),(2,a),(2,b),(3,a),(3,b)}
n(A×B) = n(A) × n(B) = 3 × 2 = 6

Properties:
1. $A \times B \neq B \times A$ (ordered pairs are ordered)
2. n(A×B) = n(A) × n(B)
3. $A \times \phi = \phi$
4. $A \times (B \cup C) = (A \times B) \cup (A \times C)$
5. If A = B, then A×A is called A squared.

Application: Used in relational databases (tables are Cartesian products of attribute domains)

Q17. Explain minor and cofactor with an example. [4M]

MINOR: For a matrix A, the minor M_{ij} of element a_{ij} is the determinant of the submatrix formed by deleting row i and column j.

Example: A = [[1,2,3],[4,5,6],[7,8,9]]
M_{11} = |5,6; 8,9| = 45-48 = -3 (delete row 1, col 1)
M_{12} = |4,6; 7,9| = 36-42 = -6 (delete row 1, col 2)
M_{13} = |4,5; 7,8| = 32-35 = -3 (delete row 1, col 3)

COFACTOR: $A_{ij} = (-1)^{(i+j)} \times M_{ij}$
$A_{11} = (-1)^2 \times (-3) = +1 \times (-3) = -3$
$A_{12} = (-1)^3 \times (-6) = -1 \times (-6) = +6$
$A_{13} = (-1)^4 \times (-3) = +1 \times (-3) = -3$

Sign pattern: [+,-,+; -,+,-; +,-,+]

Q18. What are the properties of sets? [2M]

Important Properties:
1. Commutative: $A \cup B = B \cup A$ and $A \cap B = B \cap A$
2. Associative: $A \cup (B \cup C) = (A \cup B) \cup C$ and $A \cap (B \cap C) = (A \cap B) \cap C$
3. Distributive: $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$
4. Identity: $A \cup \phi = A$ and $A \cap U = A$
5. Complement: $A \cup A' = U$ and $A \cap A' = \phi$
6. De Morgan: $(A \cup B)' = A' \cap B'$ and $(A \cap B)' = A' \cup B'$
7. Idempotent: $A \cup A = A$ and $A \cap A = A$

Q19. Solve using Cramer's Rule: x+2y+z=6, 2x+y+z=7, x+y+2z=7. [6M]

System: $x+2y+z=6$, $2x+y+z=7$, $x+y+2z=7$
Coefficient Matrix A = [[1,2,1],[2,1,1],[1,1,2]]

D = |1,2,1; 2,1,1; 1,1,2|
$= 1(1\times2-1\times1) - 2(2\times2-1\times1) + 1(2\times1-1\times1)$
$= 1(2-1) - 2(4-1) + 1(2-1)$
$= 1-6+1 = -4$

Dx = |6,2,1; 7,1,1; 7,1,2|
$= 6(2-1)-2(14-7)+1(7-7) = 6-14+0 = -8$

Dy = |1,6,1; 2,7,1; 1,7,2|
$= 1(14-7)-6(4-1)+1(14-7) = 7-18+7 = -4$

Dz = |1,2,6; 2,1,7; 1,1,7|
$= 1(7-7)-2(14-7)+6(2-1) = 0-14+6 = -8$

$x=Dx/D=-8/-4=2$, $y=Dy/D=-4/-4=1$, $z=Dz/D=-8/-4=2$
Solution: $x=2$, $y=1$, $z=2$

Q20. What is logical equivalence? State laws of logical equivalence. [4M]

Logical Equivalence: Statements P and Q are logically equivalent ($P \equiv Q$) if they have identical truth values for all possible combinations of truth values of their variables.
Verified using truth tables — columns must be identical.

Important Laws:
1. Double Negation: $\neg(\neg p) \equiv p$
2. De Morgan's: $\neg(p \wedge q) \equiv \neg p \vee \neg q$ and $\neg(p \vee q) \equiv \neg p \wedge \neg q$
3. Commutative: $p \wedge q \equiv q \wedge p$; $p \vee q \equiv q \vee p$
4. Associative: $p \wedge (q \wedge r) \equiv (p \wedge q) \wedge r$
5. Distributive: $p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$
6. Identity: $p \wedge T \equiv p$; $p \vee F \equiv p$
7. Domination: $p \vee T \equiv T$; $p \wedge F \equiv F$
8. Idempotent: $p \wedge p \equiv p$; $p \vee p \equiv p$
9. Absorption: $p \vee (p \wedge q) \equiv p$; $p \wedge (p \vee q) \equiv p$